

PY1010 — Physics for Biological and Chemical Sciences

Summer 2023 Examination · Paper 1 · Full Marking Scheme

Module	PY1010 — Physics for Biological and Chemical Sciences
Internal examiners	Prof. P. Callanan, Prof. A. Ruth
Institution	University College Cork
Session	Summer 2023, Paper 1
Duration	3 hours (+ 15 minutes reading time)
Structure	Six questions, each given as an either/or choice (A or B). Answer one option of each of the six questions and all parts of that option. Marks for each part are shown in square brackets.
Materials	Non-programmable calculator permitted; constants on the paper's "useful data" page

How to use this scheme. Every answer is worked out in small numbered steps, with a "Rules you need" box listing the formulas used. Nothing is skipped, so you can see where every number comes from: every unit conversion, every substitution, every check. Read the answer first, then follow the steps.

Coverage note. This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

Source basis. Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, waves and optics). No textbook or outside material is used, except where a highlighted note says the lecture notes don't cover something (Q5(A)(a) and Q6(B)).

Own-figure rule. Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part.

Numbers. $g = 9.81 \text{ m s}^{-2}$ and other constants from the paper's useful-data page. Intermediate values are kept unrounded; final answers are given to 2–3 significant figures. Answers differing only through rounding get full marks.

Study-Unit

Question 1 · Option A

Volumes, areas and mass of a year's rainfall over Ireland — 10 marks

Given: Annual rainfall 1151 mm, assumed over all of Ireland; area of Ireland 84 421 km²; raindrop radius 1 mm; $\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$.

1(A)(a)

[3 marks]

What volume of rain fell over every square kilometre last year? Express your answer in m³.

ANSWER

$1.151 \times 10^6 \text{ m}^3 \text{ per km}^2$

RULES YOU NEED

- Volume of a flat layer = area $\times$ depth
- 1 mm = 10^{-3} m
- 1 km = 10^3 m, so 1 km² = $(10^3)^2 \text{ m}^2 = 10^6 \text{ m}^2$

WORKING — STEP BY STEP

1 Picture the rain as a flat layer

If all the year's rain lay on the ground without draining away, it would form a layer 1151 mm deep. Its volume is area $\times$ depth.

2 Convert the depth to metres

$$1151 \text{ mm} = 1151 \times 10^{-3} \text{ m} = 1.151 \text{ m}$$

3 Convert 1 km² to m²

A square 1 km on each side is 1000 m $\times$ 1000 m:

$$1 \text{ km}^2 = 10^3 \text{ m} \times 10^3 \text{ m} = 10^6 \text{ m}^2$$

4 Multiply

$$V = 10^6 \text{ m}^2 \times 1.151 \text{ m} = 1.151 \times 10^6 \text{ m}^3$$

Marks	Awarded for
1	Depth in metres (1.151 m).
1	1 km ² = 10 ⁶ m ² .
1	V = 1.151 $\times$ 10 ⁶ m ³ .

1(A)(b)**[2 marks]**

If the total area of Ireland is 84 421 km², what total volume of rain fell on the country last year?

ANSWER

$$V_{\text{total}} \approx 9.7 \times 10^{10} \text{ m}^3$$

RULES YOU NEED

- Total = (volume per km²) × (number of km²)

WORKING — STEP BY STEP**1 Scale up from one square kilometre**

Each km² gets $1.151 \times 10^6 \text{ m}^3$, and there are 84 421 km²:

$$V_{\text{total}} = 84\,421 \times 1.151 \times 10^6 \text{ m}^3$$

2 Calculate

$84\,421 \times 1.151 = 97\,169$, so:

$$V_{\text{total}} = 97\,169 \times 10^6 \text{ m}^3 = 9.72 \times 10^{10} \text{ m}^3$$

Marks	Awarded for
1	Multiplies by the area in km ² .
1	$\approx 9.7 \times 10^{10} \text{ m}^3$ (own-figure from (a) accepted).

1(A)(c)**[3 marks]**

If each raindrop is a sphere of radius 1 mm, how many raindrops fell on Ireland last year?

ANSWER

$N \approx 2.3 \times 10^{19}$ raindrops

RULES YOU NEED

- Volume of a sphere: $V = (4/3) \pi r^3$
- Number of drops = total volume $\div$ volume of one drop

WORKING — STEP BY STEP**1 Radius in metres**

$$r = 1 \text{ mm} = 10^{-3} \text{ m}$$

2 Volume of one drop

$$r^3 = (10^{-3})^3 = 10^{-9} \text{ m}^3$$

$$V_{\text{drop}} = (4/3) \times \pi \times 10^{-9} = 4.19 \times 10^{-9} \text{ m}^3$$

3 Divide the total volume by one drop

$$N = 9.72 \times 10^{10} \div 4.19 \times 10^{-9}$$

4 Calculate with powers of ten

$$9.72 \div 4.19 = 2.32, \text{ and } 10^{10} \div 10^{-9} = 10^{19}:$$

$$N = 2.3 \times 10^{19} \text{ drops}$$

Marks	Awarded for
1	Sphere volume with r in metres.
1	$N = V_{\text{total}}/V_{\text{drop}}$.
1	$\approx 2.3 \times 10^{19}$.

1(A)(d)**[2 marks]**

The density of water is 1000 kg m^{-3} . What was the total mass of rain water falling on Ireland last year?

ANSWER

$$m \approx 9.7 \times 10^{13} \text{ kg}$$

RULES YOU NEED

- $m = \rho V$

WORKING — STEP BY STEP**1 Use $m = \rho V$ with the volume from (b)**

$$m = 1000 \text{ kg/m}^3 \times 9.72 \times 10^{10} \text{ m}^3$$

2 Calculate

$$m = 9.72 \times 10^{13} \text{ kg}$$

Marks	Awarded for
1	$m = \rho V$.
1	$\approx 9.7 \times 10^{13} \text{ kg}$.

Question 1(A) Total: 10 Marks

Question 1 · Option B

Constant acceleration: Usain Bolt's 200 m — 10 marks

Given: 200 m in 19.19 s (Berlin, 2009).

1(B)(a)

[3 marks]

If Usain Bolt crossed the finish line after constant acceleration from the start, what would this acceleration have been, in m s^{-2} ?

ANSWER

$$a \approx 1.09 \text{ m s}^{-2}$$

RULES YOU NEED

- Constant acceleration from rest: $x = v_0 t + \frac{1}{2} a t^2$, with $v_0 = 0$

WORKING — STEP BY STEP

1 List what you know

He starts from rest ($v_0 = 0$), covers $x = 200 \text{ m}$, in $t = 19.19 \text{ s}$. We want a .

2 Pick the equation with x , t and a

$$x = v_0 t + \frac{1}{2} a t^2 \Rightarrow 200 = 0 + \frac{1}{2} a (19.19)^2$$

3 Rearrange for a

$$a = 2x / t^2 = 2 \times 200 \div 19.19^2$$

4 Calculate

$$19.19^2 = 368.26:$$

$$a = 400 \div 368.26 = 1.09 \text{ m s}^{-2}$$

Marks	Awarded for
1	$v_0 = 0$ and $x = v_0 t + \frac{1}{2} a t^2$ chosen.
1	Rearranged for a .
1	$a \approx 1.09 \text{ m s}^{-2}$.

1(B)(b)**[3 marks]**

Using the acceleration from (a), what would his speed have been as he crossed the finish line?

ANSWER

$$v \approx 20.8 \text{ m s}^{-1}$$

RULES YOU NEED

- $v = v_0 + a t$
- Check: $v^2 = v_0^2 + 2a x$

WORKING — STEP BY STEP**1 Use $v = v_0 + at$**

$$v = 0 + 1.086 \times 19.19$$

2 Calculate

$$v = 20.8 \text{ m/s}$$

3 Check with a second equation

$$v^2 = 2 \times 1.086 \times 200 = 434.5 \Rightarrow v = \sqrt{434.5} = 20.8 \text{ m/s } \checkmark$$

Marks	Awarded for
1	Suitable equation ($v = u + at$ or $v^2 = u^2 + 2as$).
1	Correct substitution.
1	$v \approx 20.8 \text{ m s}^{-1}$ (own-figure accepted).

1(B)(c)**[4 marks]**

He starts the last 200 m leg of a relay at 5 m s^{-1} and accelerates at 1 m s^{-2} .
What is his final speed, and how long does the 200 m take?

ANSWER

$$v \approx 20.6 \text{ m s}^{-1}; t \approx 15.6 \text{ s}$$

RULES YOU NEED

- $v^2 = v_0^2 + 2a x$ (no time needed)
- $v = v_0 + a t$, so $t = (v - v_0)/a$

WORKING — STEP BY STEP**1 List what you know**

$v_0 = 5 \text{ m/s}$, $a = 1 \text{ m/s}^2$, $x = 200 \text{ m}$. Unknowns: v and t .

2 Final speed first (no time needed)

$$v^2 = 5^2 + 2 \times 1 \times 200 = 25 + 400 = 425$$

$$v = \sqrt{425} = 20.6 \text{ m/s}$$

3 Now the time

$$t = (v - v_0) / a = (20.6 - 5) \div 1 = 15.6 \text{ s}$$

4 Check with $x = v_0 t + \frac{1}{2} a t^2$

$$5 \times 15.6 + \frac{1}{2} \times 1 \times 15.6^2 = 78 + 122 = 200 \text{ m } \checkmark$$

Marks	Awarded for
1	$v^2 = u^2 + 2as$.
1	$v \approx 20.6 \text{ m s}^{-1}$.
1	$v = u + at$ rearranged for t .
1	$t \approx 15.6 \text{ s}$.

Question 1(B) Total: 10 Marks

Question 2 · Option A

Newton's law of gravitation, mass and weight, a diving kingfisher — 10 marks

Given: Kingfisher on a branch 4 m above the pond; dives using only gravity; negligible air resistance.

2(A)(a)

[2 marks]

State and explain Newton's law of Gravitation. Give the SI units of the quantities in the law.

ANSWER

$F = Gm_1m_2/r^2$: attractive, $\propto$ product of masses, $\propto 1/\text{distance}^2$

RULES YOU NEED

- $F = G m_1 m_2 / r^2$
- $G = 6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$

WORKING — STEP BY STEP

1 The law

$$F = G m_1 m_2 / r^2$$

2 What it says

Any two masses attract each other. The force is proportional to the product of the masses, and inversely proportional to the square of the distance between their centres (double the distance $\rightarrow$ a quarter of the force).

3 Units

F in N; m_1, m_2 in kg; r in m; $G = 6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ (the universal gravitational constant).

Marks	Awarded for
1	Law stated and explained.
1	SI units of F, m, r and G.

2(A)(b)(i)**[1 mark]**

How are the mass and the weight of an object related? Explain.

ANSWER

$$w = mg$$

RULES YOU NEED

- $w = m g$

WORKING — STEP BY STEP**1 The relation**

$$w = m g$$

2 Explain

Weight is the gravitational force on a mass. Mass (kg) is the amount of matter and doesn't change; weight (N) depends on the local g .

Marks**Awarded for****1** $w = mg$, with explanation.

2(A)(b)(ii)**[1 mark]**

You step from the ground onto a chair. Does your weight increase, decrease, or remain the same?

ANSWER

It decreases, but only by a negligible amount (r increases slightly, so $g = GM_E/r^2$ falls)

RULES YOU NEED

- $g = G M_E / r^2$ (r = distance from the Earth's centre)

WORKING — STEP BY STEP**1 What changes on the chair?**

You are about 0.5 m further from the Earth's centre, so r gets slightly bigger.

2 Effect on g

$g = GM_E/r^2$, so a bigger r means a slightly smaller g, and your weight mg is slightly smaller.

3 How big is the change?

Tiny: 0.5 m compared with $r \approx 6\,378\,000$ m. Your weight **decreases**, but by a negligible amount.

Marks	Awarded for
1	Decreases (negligibly), explained via $g \propto 1/r^2$.

2(A)(c)(i)**[3 marks]**

How long does it take the kingfisher to reach the prey at the water surface?

ANSWER

$t \approx 0.90 \text{ s}$

RULES YOU NEED

- Free fall from rest: $x = \frac{1}{2} g t^2$, so $t = \sqrt{(2h/g)}$

WORKING — STEP BY STEP**1 Set up free fall**

It starts from rest ($v_0 = 0$) and only gravity acts: $a = g = 9.81 \text{ m/s}^2$. It falls $h = 4 \text{ m}$.

2 Use $x = \frac{1}{2}gt^2$

$$4 = \frac{1}{2} \times 9.81 \times t^2$$

3 Rearrange for t

$$t^2 = 2 \times 4 \div 9.81 = 0.8155 \Rightarrow t = \sqrt{0.8155}$$

4 Calculate

$$t = 0.90 \text{ s}$$

Marks	Awarded for
1	Free fall with $u = 0$, $a = g$.
1	$t = \sqrt{(2h/g)}$.
1	$t \approx 0.90 \text{ s}$.

2(A)(c)(ii)**[3 marks]**

How fast is the kingfisher when it hits the prey?

ANSWER

$v \approx 8.9 \text{ m s}^{-1}$

RULES YOU NEED

- $v = \sqrt{2gh}$ (free fall from rest)
- or $v = g t$

WORKING — STEP BY STEP**1 Use $v = \sqrt{2gh}$**

$$v = \sqrt{2 \times 9.81 \times 4} = \sqrt{78.48}$$

2 Calculate

$$v = 8.86 \text{ m/s}$$

3 Check with $v = g t$

$$v = 9.81 \times 0.903 = 8.86 \text{ m/s } \checkmark$$

Marks	Awarded for
1	$v = \sqrt{2gh}$ or $v = g t$.
1	Correct substitution.
1	$v \approx 8.9 \text{ m s}^{-1}$.

Question 2(A) Total: 10 Marks

Question 2 · Option B

Archimedes' principle: a cube submerged in a fluid — 10 marks

Given: Cube of side 20.00 cm fully immersed; pressure 105 kPa at its top face, 107 kPa at its bottom face; air pressure 99 800 Pa.

2(B)(a)

[2 marks]

State Archimedes' Principle.

ANSWER

Buoyant force = weight of the fluid displaced

RULES YOU NEED

- $F_{\text{buoy}} = \text{weight of fluid displaced}$

WORKING — STEP BY STEP

1 The statement

When a body is completely or partially immersed in a fluid, the fluid exerts an **upward force** on it equal to the **weight of the fluid displaced**.

Marks	Awarded for
1	Upward force on an immersed body.
1	Equal to the weight of fluid displaced.

2(B)(b)(i)**[2 marks]**

What is the net upwards (buoyant) force acting on the cube?

ANSWER

$$F_{\text{buoy}} = 80 \text{ N (upwards)}$$

RULES YOU NEED

- Force = pressure × area
- 1 kPa = 1000 Pa

WORKING — STEP BY STEP**1 Why is there an upward force?**

The pressures on the four sides cancel out. The bottom face is deeper, so it is pushed up harder (107 kPa) than the top face is pushed down (105 kPa).

2 Area of one face

$$A = (0.20 \text{ m})^2 = 0.04 \text{ m}^2$$

3 Pressure difference

$$\Delta P = 107\,000 - 105\,000 = 2000 \text{ Pa}$$

4 Net force

$$F = \Delta P \times A = 2000 \times 0.04 = 80 \text{ N (upwards)}$$

Marks	Awarded for
1	Uses the pressure difference × face area.
1	80 N.

2(B)(b)(ii)**[2 marks]**

Use Archimedes' Principle to calculate the mass of the displaced fluid.

ANSWER

$m_{\text{fl}} \approx 8.2 \text{ kg}$

RULES YOU NEED

- Buoyant force = weight of displaced fluid = $m_{\text{fl}} g$

WORKING — STEP BY STEP**1 Set the buoyant force equal to the fluid's weight**

$$80 \text{ N} = m_{\text{fl}} \times 9.81$$

2 Solve for the mass

$$m_{\text{fl}} = 80 \div 9.81 = 8.15 \text{ kg}$$

Marks	Awarded for
1	$F = m_{\text{fl}} g.$
1	$\approx 8.2 \text{ kg}.$

2(B)(b)(iii)**[2 marks]**

What is the density of the displaced fluid?

ANSWER

$$\rho \approx 1020 \text{ kg m}^{-3}$$

RULES YOU NEED

- $\rho = m / V$
- Displaced volume = volume of the cube (fully immersed)

WORKING — STEP BY STEP**1 Volume displaced**

$$V = (0.20 \text{ m})^3 = 0.008 \text{ m}^3 = 8.0 \times 10^{-3} \text{ m}^3$$

2 Density

$$\rho = 8.155 \text{ kg} \div 0.008 \text{ m}^3 = 1019 \text{ kg m}^{-3}$$

3 Sense checkJust above water's 1000 kg m^{-3} , e.g. salty water: sensible ✓

Marks	Awarded for
1	$V = 8 \times 10^{-3} \text{ m}^3$.
1	$\rho \approx 1020 \text{ kg m}^{-3}$.

2(B)(b)(iv)**[2 marks]**

What is the depth d of the cube below the surface of the fluid? Air pressure was 99 800 Pa.

ANSWER

$$d \approx 0.52 \text{ m}$$

RULES YOU NEED

- Pressure at depth d : $P = P_{\text{atm}} + \rho g d$

WORKING — STEP BY STEP**1 Write the pressure at the top face**

$$P_{\text{top}} = P_{\text{atm}} + \rho g d$$

2 Rearrange for d

$$d = (P_{\text{top}} - P_{\text{atm}}) / (\rho g)$$

3 Put the numbers in

$$d = (105\,000 - 99\,800) \div (1019 \times 9.81) = 5200 \div 10\,000 = 0.52 \text{ m}$$

Marks	Awarded for
1	$P = P_{\text{atm}} + \rho g d$.
1	$d \approx 0.52 \text{ m}$.

Question 2(B) Total: 10 Marks

Question 3 · Option A

Specific heat, latent heat of fusion, defrosting a freezer — 10 marks

Given: Hairdryer 1 kW; 1 kg of ice at $-10\text{ }^{\circ}\text{C}$; $c_{\text{ice}} = 2.22 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$; $L_f = 3.35 \times 10^5 \text{ J kg}^{-1}$.

3(A)(a)

[2 marks]

Define (i) specific heat capacity and (ii) heat of fusion.

ANSWER

$c = Q/(m\Delta T) \text{ [J kg}^{-1} \text{ K}^{-1}\text{]}; L_f = Q/m \text{ for melting at constant T [J kg}^{-1}\text{]}$

RULES YOU NEED

- $Q = m c \Delta T$
- $Q = m L$ (phase change, no temperature change)

WORKING — STEP BY STEP

1 Specific heat capacity

The heat needed per unit mass to raise a substance's temperature by 1 K: $c = Q/(m\Delta T)$, unit $\text{J kg}^{-1} \text{ K}^{-1}$.

2 Heat of fusion

The heat needed per unit mass to change a substance from solid to liquid at its melting point, **without** changing its temperature: $L_f = Q/m$, unit J kg^{-1} .

Marks	Awarded for
1	Specific heat capacity defined.
1	Heat of fusion defined (solid $\rightarrow$ liquid at constant T).

3(A)(b)(i)**[2 marks]**

Calculate the heat needed to raise the temperature of the ice to 0 °C.

ANSWER

$$Q_1 = 2.22 \times 10^4 \text{ J (22.2 kJ)}$$

RULES YOU NEED

- $Q = m c \Delta T$

WORKING — STEP BY STEP**1 Temperature change**

$$\Delta T = 0 - (-10) = 10 \text{ °C} = 10 \text{ K}$$

2 Apply $Q = mc\Delta T$

$$Q_1 = 1 \times 2220 \times 10 = 22\,200 \text{ J} = 2.22 \times 10^4 \text{ J}$$

Marks	Awarded for
1	$Q = mc\Delta T$ with $\Delta T = 10 \text{ K}$.
1	22.2 kJ.

3(A)(b)(ii)**[2 marks]**

Calculate the heat required to convert the ice at 0 °C to water at the same temperature.

ANSWER

$$Q_2 = 3.35 \times 10^5 \text{ J (335 kJ)}$$

RULES YOU NEED

- $Q = m L_f$

WORKING — STEP BY STEP**1 Melting at constant temperature uses latent heat**

$$Q_2 = m L_f = 1 \times 3.35 \times 10^5 = 3.35 \times 10^5 \text{ J}$$

Marks	Awarded for
1	$Q = mL_f$.
1	335 kJ.

3(A)(b)(iii)**[4 marks]**

If 50% of the heat from the hairdryer heats the ice, how long will it take to melt all the ice?

ANSWER

$t \approx 714 \text{ s}$ (about 12 minutes)

RULES YOU NEED

- Total heat = warming + melting
- Power = energy $\div$ time, so $t = Q / P$
- 1 kW = 1000 W = 1000 J/s

WORKING — STEP BY STEP**1 Total heat needed**

$$Q = Q_1 + Q_2 = 22\,200 + 335\,000 = 357\,200 \text{ J}$$

2 Useful power

Only 50% of the 1000 W reaches the ice:

$$P = 0.5 \times 1000 = 500 \text{ W}$$

3 Time

$$t = Q / P = 357\,200 \div 500 = 714 \text{ s}$$

4 In minutes

$$714 \div 60 \approx 11.9 \text{ min}$$

Marks	Awarded for
1	Total heat $Q_1 + Q_2$.
1	Useful power = 500 W.
1	$t = Q/P$.
1	$\approx 714 \text{ s}$.

Question 3(A) Total: 10 Marks

Question 3 · Option B

Ideal gas law $PV = NkT$, Boyle's and Charles' laws, the spy balloon — 10 marks

Given: Balloon radius 10 m at 20 km: $T = -58\text{ }^{\circ}\text{C}$, $P = 5000\text{ Pa}$. At 5 km: $T = -13\text{ }^{\circ}\text{C}$, $P = 50\,000\text{ Pa}$. Inside always matches outside.

3(B)(a)(i)

[2 marks]

Write down the SI units for P and T in $PV = NkT$.

ANSWER

P in Pa (N m^{-2}); T in K

RULES YOU NEED

- Pressure = force / area
- Gas laws need absolute temperature

WORKING — STEP BY STEP

1 Pressure

Force per area: N m^{-2} , called the **pascal (Pa)**.

2 Temperature

Must be the absolute temperature, in **kelvin (K)**.

Marks	Awarded for
1	Pa.
1	K.

3(B)(a)(ii)**[1 mark]**

Under what conditions will this expression lead to Boyle's Law?

ANSWER

Fixed amount of gas (N constant) at constant T $\Rightarrow$ $PV = \text{constant}$

RULES YOU NEED

- Boyle's law: $PV = \text{constant}$ ($P \propto 1/V$)

WORKING — STEP BY STEP**1 Hold things fixed**

Fixed amount of gas (N constant) at constant temperature T. Then NkT is constant, so $PV = \text{constant}$: Boyle's law.

Marks**Awarded for****1**

N and T constant.

3(B)(a)(iii)**[1 mark]**

Under what conditions will this expression lead to Charles' Law?

ANSWER

Fixed amount of gas at constant P $\Rightarrow$ $V/T = \text{constant}$

RULES YOU NEED

- Charles' law: $V/T = \text{constant}$

WORKING — STEP BY STEP**1 Hold things fixed**

Fixed amount of gas at constant pressure P. Then $V = (Nk/P)T$, so $V/T = \text{constant}$: Charles' law.

Marks**Awarded for****1**

N and P constant.

3(B)(b)(i)**[3 marks]**

What is the number of gas molecules in the balloon?

ANSWER

$N \approx 7.0 \times 10^{27}$

RULES YOU NEED

- $PV = NkT$, so $N = PV/(kT)$
- Sphere volume $V = (4/3)\pi r^3$
- $T_K = T_{°C} + 273.15$
- $k = 1.381 \times 10^{-23} \text{ J K}^{-1}$

WORKING — STEP BY STEP**1 Volume of the balloon**

$$V = (4/3) \pi (10)^3 = (4/3) \pi \times 1000 = 4189 \text{ m}^3$$

2 Temperature in kelvin

$$T = -58 + 273.15 = 215.15 \text{ K}$$

3 Rearrange $PV = NkT$

$$N = P V / (k T)$$

4 Top and bottom separately

$$PV = 5000 \times 4189 = 2.094 \times 10^7$$

$$kT = 1.381 \times 10^{-23} \times 215.15 = 2.971 \times 10^{-21}$$

5 Divide

$$N = 2.094 \times 10^7 \div 2.971 \times 10^{-21} = 7.0 \times 10^{27}$$

Marks	Awarded for
1	Sphere volume and T in K.
1	$N = PV/(kT)$.
1	$\approx 7.0 \times 10^{27}$.

3(B)(b)(ii)**[3 marks]**

Calculate the radius of the balloon at 5 km.

ANSWER

$r \approx 4.9 \text{ m}$

RULES YOU NEED

- Fixed amount of gas: $P_1 V_1 / T_1 = P_2 V_2 / T_2$
- $r = (3V / 4\pi)^{1/3}$

WORKING — STEP BY STEP**1 New temperature in kelvin**

$$T_2 = -13 + 273.15 = 260.15 \text{ K}$$

2 Rearrange for V_2

$$V_2 = V_1 \times (P_1 / P_2) \times (T_2 / T_1)$$

3 Put the numbers in

$$V_2 = 4189 \times (5000 / 50\,000) \times (260.15 / 215.15) = 4189 \times 0.1 \times 1.209 = 506 \text{ m}^3$$

Pressure $\times 10$ squashes it a lot; the small warm-up expands it a little.**4 Turn the volume back into a radius**

$$r^3 = 3V / (4\pi) = 3 \times 506 \div 12.57 = 120.9$$

$$r = \sqrt[3]{120.9} = 4.9 \text{ m}$$

Marks	Awarded for
1	$P_1 V_1 / T_1 = P_2 V_2 / T_2$ with T in K.
1	$V_2 \approx 506 \text{ m}^3$.
1	$r \approx 4.9 \text{ m}$.

Question 3(B) Total: 10 Marks

Question 4 · Option A

$v = f\lambda$, decibels, intensity and distance: hearing a pin drop — 10 marks

Given: A pin releases 10^{-11} J of sound energy over 1 ms, radiated equally in all directions; hearing threshold $I = 10^{-12} \text{ W m}^{-2}$.

4(A)(a)

[1 mark]

Write down the formula relating the frequency, wavelength and speed of a wave.

ANSWER

$$v = f\lambda$$

RULES YOU NEED

- $v = f\lambda$

WORKING — STEP BY STEP

1 The formula

$$v = f\lambda$$

v in m s^{-1} , f in Hz, λ in m. It holds for all waves.

Marks

Awarded for

1

$v = f\lambda$.

4(A)(b)**[1 mark]**

Write down the formula relating a sound intensity in W m^{-2} to the sound intensity level in dB.

ANSWER

$$\text{dB} = 10 \log_{10}(I/I_0), \text{ with } I_0 = 10^{-12} \text{ W m}^{-2}$$

RULES YOU NEED

- $I_0 = 10^{-12} \text{ W m}^{-2}$ (threshold of hearing)

WORKING — STEP BY STEP**1 The formula**

$$\text{dB} = 10 \log_{10}(I / I_0)$$

2 The terms

I is the sound intensity (W m^{-2}); $I_0 = 10^{-12} \text{ W m}^{-2}$ is the quietest sound a human can hear. Lecture example: $I = 10^{-10}$ gives $10 \log(100) = 20 \text{ dB}$.

Marks**Awarded for****1**Formula with I_0 identified.

4(A)(c)(i)**[2 marks]**What is the rate, L , at which this energy is radiated, in W?**ANSWER**

$L = 1 \times 10^{-8} \text{ W}$

RULES YOU NEED

- Power = energy $\div$ time
- 1 ms = 10^{-3} s

WORKING — STEP BY STEP**1 Time in seconds**

$$t = 1 \text{ ms} = 10^{-3} \text{ s}$$

2 Divide energy by time

$$L = 10^{-11} \text{ J} \div 10^{-3} \text{ s} = 10^{-11 - (-3)} = 10^{-8} \text{ W}$$

Marks	Awarded for
1	$L = E/t$ with $1 \text{ ms} = 10^{-3} \text{ s}$.
1	10^{-8} W .

4(A)(c)(ii)**[3 marks]**

At what maximum distance d from the pin could you hear it hit the floor? Use $L / (4\pi d^2) = I$.

ANSWER

$$d \approx 28 \text{ m}$$

RULES YOU NEED

- Isotropic source: $I = L / (4\pi d^2)$
- Rearranged: $d = \sqrt{L / (4\pi I)}$

WORKING — STEP BY STEP**1 Why this formula?**

The sound spreads out equally in all directions, over a sphere of area $4\pi d^2$. The intensity is the power spread over that area.

2 Rearrange for d

$$4\pi d^2 = L / I \Rightarrow d = \sqrt{L / (4\pi I)}$$

3 Put the numbers in

$$L / (4\pi I) = 10^{-8} \div (4\pi \times 10^{-12}) = 10^{-8} \div 1.257 \times 10^{-11} = 795.8 \text{ m}^2$$

4 Square root

$$d = \sqrt{795.8} = 28.2 \text{ m}$$

Marks	Awarded for
1	Rearranged for d .
1	Correct substitution.
1	$d \approx 28 \text{ m}$.

4(A)(c)(iii)**[3 marks]**

If the frequency is 10 kHz and the wavelength 3 cm, how long does the sound take to travel from the pin to your ear? Use (ii).

ANSWER

$$t \approx 0.094 \text{ s (94 ms)}$$

RULES YOU NEED

- $v = f \lambda$
- $t = d / v$
- 1 kHz = 10^3 Hz; 1 cm = 10^{-2} m

WORKING — STEP BY STEP**1 Convert to SI**

$$f = 10 \text{ kHz} = 10^4 \text{ Hz}; \lambda = 3 \text{ cm} = 0.03 \text{ m}$$

2 Speed of the sound

$$v = f \lambda = 10^4 \times 0.03 = 300 \text{ m/s}$$

3 Time over the distance from (ii)

$$t = d / v = 28.2 \div 300 = 0.094 \text{ s}$$

Marks	Awarded for
1	$v = f\lambda = 300 \text{ m s}^{-1}$, with units converted.
1	$t = d/v$.
1	$\approx 0.094 \text{ s}$ (own-figure from (ii) accepted).

Question 4(A) Total: 10 Marks

Question 4 · Option B

Coulomb's law, the electric field, a charged water drop — 10 marks

Given: Water drop carrying 100 times the charge of an electron, held up by $E = 10^5 \text{ V m}^{-1}$.

4(B)(a)

[1 mark]

State Coulomb's law for the interaction between two electric charges.

ANSWER

$$F = Qq/(4\pi\epsilon_0 r^2)$$

RULES YOU NEED

- $F = Q q / (4\pi \epsilon_0 r^2)$

WORKING — STEP BY STEP

1 The law

$$F = Q q / (4\pi \epsilon_0 r^2)$$

2 What it says

The force between two point charges is proportional to the product of the charges and inversely proportional to the square of their distance apart. Like charges repel; unlike charges attract.

Marks	Awarded for
1	Law stated.

4(B)(b)**[2 marks]**

Define both the direction and magnitude of an electric field E at a point.

ANSWER

$E = F/q$; direction = direction of force on a positive charge

RULES YOU NEED

- $F = qE$, so $E = F/q$

WORKING — STEP BY STEP**1 Magnitude**

Place a small test charge q at the point. E is the force on it per unit charge: $E = F/q$ (unit $\text{N C}^{-1} = \text{V m}^{-1}$).

2 Direction

The direction of the force a **positive** test charge would feel there.

Marks	Awarded for
1	Magnitude $E = F/q$.
1	Direction of force on a positive test charge.

4(B)(c)**[2 marks]**

An electron, released from rest, moves left to right. What is the direction of E ?
Does its electric potential energy increase or decrease?

ANSWER**E points right → left; potential energy decreases****RULES YOU NEED**

- $F = qE$: for a negative q , the force is opposite to E
- Free charges move from higher to lower electric PE

WORKING — STEP BY STEP**1 Direction of the force**

The electron speeds up to the right, so the force on it points right.

2 Direction of E

The electron is negative, so E points the opposite way to the force: **right to left**.

3 Potential energy

Released from rest, it moves by itself, from higher to lower PE, like a ball rolling downhill. Its PE **decreases** and turns into kinetic energy.

Marks	Awarded for
1	E right to left.
1	PE decreases.

4(B)(d)(i)**[1 mark]**

What is the direction of E?

ANSWER**E points vertically downwards****RULES YOU NEED**

- For a negative charge, the electric force is opposite to E

WORKING — STEP BY STEP**1 Which way must the electric force point?**It must hold the drop up against gravity, so it points **up**.**2 The drop is negative**It carries electron charge, so the force is opposite to E. E points **down**.

Marks	Awarded for
1	Downwards.

4(B)(d)(ii)**[4 marks]**

What is the mass of the drop?

ANSWER

$$m \approx 1.6 \times 10^{-13} \text{ kg}$$

RULES YOU NEED

- Balanced forces: electric force = weight, $qE = mg$
- $e = 1.602 \times 10^{-19} \text{ C}$

WORKING — STEP BY STEP**1 Charge on the drop**

$$q = 100 \times 1.602 \times 10^{-19} = 1.602 \times 10^{-17} \text{ C}$$

2 The drop is still, so the forces balance

$$q E = m g$$

3 Rearrange for m

$$m = q E / g$$

4 Put the numbers in

$$m = 1.602 \times 10^{-17} \times 10^5 \div 9.81 = 1.602 \times 10^{-12} \div 9.81 = 1.63 \times 10^{-13} \text{ kg}$$

Marks	Awarded for
1	$q = 100e$.
1	Force balance $qE = mg$.
1	Rearranged for m .
1	$\approx 1.6 \times 10^{-13} \text{ kg}$.

Question 4(B) Total: 10 Marks

Question 5 · Option A

Resistors in parallel, Ohm's law, ohmic and non-ohmic resistors — 10 marks

Given: Resistors A ($100\ \Omega$) and B in parallel across a 200 V battery; 1 A flows through B.

5(A)(a)

[2 marks]

Draw a simple circuit with two resistors connected in parallel, and write down an expression for their equivalent resistance.

ANSWER

$$1/R_{\text{eq}} = 1/R_1 + 1/R_2$$

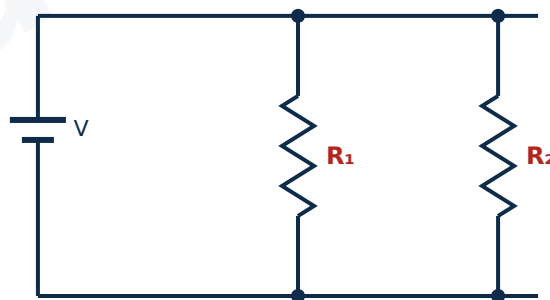
RULES YOU NEED

- Parallel: each resistor is connected across the same two points
- $1/R_{\text{eq}} = 1/R_1 + 1/R_2$

WORKING — STEP BY STEP

1 Draw the circuit

Connect both resistors side by side between the same two wires, so each one sits directly across the battery.



Both resistors are connected across the same two

Fig. 1 — Two resistors in parallel across a battery: each has the full battery voltage across it.

2 Equivalent resistance

$$1/R_{\text{eq}} = 1/R_1 + 1/R_2$$

The combination has less resistance than either resistor alone, since the current has two paths.

Note on the lecture material: The PY1010 lecture notes don't cover combining resistors in parallel (or in series). The equivalent-resistance formula here comes from outside the lecture material.

Marks	Awarded for
1	Correct parallel circuit diagram.
1	Equivalent resistance expression.

5(A)(b)**[1 mark]**

Write down Ohm's Law.

ANSWER

$$V = IR$$

RULES YOU NEED

- $V = IR$

WORKING — STEP BY STEP**1 The law**

$$V = IR$$

The current through a conductor is proportional to the voltage across it, with a constant resistance R .

Marks	Awarded for
1	$V = IR$.

5(A)(c)(i)**[1 mark]**

What is the voltage drop across resistor B?

ANSWER

$V_B = 200 \text{ V}$

RULES YOU NEED

- Each parallel branch is connected across the battery terminals

WORKING — STEP BY STEP**1 Look at the connections**

B is connected directly across the two battery terminals, so it has the full battery voltage across it.

Marks	Awarded for
-------	-------------

1	200 V.
---	--------

5(A)(c)(ii)**[1 mark]**

What is the resistance of B?

ANSWER

$R_B = 200 \Omega$

RULES YOU NEED

- $R = V / I$

WORKING — STEP BY STEP**1 Use $V = IR$**

$$R_B = 200 \text{ V} \div 1 \text{ A} = 200 \Omega$$

Marks	Awarded for
-------	-------------

1	200 Ω .
---	----------------

5(A)(c)(iii)**[2 marks]**

What current flows through resistor A?

ANSWER

$$I_A = 2 \text{ A}$$

RULES YOU NEED

$$\bullet I = V / R$$

WORKING — STEP BY STEP**1 Voltage across A**

A is also directly across the battery, so it has 200 V too.

2 Current

$$I_A = 200 \text{ V} \div 100 \, \Omega = 2 \text{ A}$$

Marks	Awarded for
1	200 V across A, $I = V/R$.
1	2 A.

5(A)(d)**[3 marks]**

When $V = 10\text{ V}$, $I = 1\text{ A}$; when $V = 20\text{ V}$, $I = 1.5\text{ A}$. Is R ohmic or non-ohmic? Why?

ANSWER

Non-ohmic: R changes from $10\ \Omega$ to $13.3\ \Omega$ (I is not proportional to V)

RULES YOU NEED

- $R = V / I$
- Ohmic: R stays constant, so $I \propto V$

WORKING — STEP BY STEP**1 Resistance at 10 V**

$$R_1 = 10 \div 1 = 10\ \Omega$$

2 Resistance at 20 V

$$R_2 = 20 \div 1.5 = 13.3\ \Omega$$

3 What would an ohmic resistor do?

Its R would stay at $10\ \Omega$, so doubling V to 20 V would double I to 2 A . Here I only reaches 1.5 A .

4 Conclusion

R changes, so the resistor is **non-ohmic**.

Note on the lecture material: The lecture notes use $V = IR$ throughout but don't name "ohmic" and "non-ohmic" resistors. The answer applies $V = IR$ with R constant.

Marks	Awarded for
1	R at both settings.
1	States R is not constant / I not $\propto V$.
1	Non-ohmic.

Question 5(A) Total: 10 Marks

Question 5 · Option B

Lorentz force on a moving charge; electric and magnetic forces combined — 10 marks

Given: Positive charge $q = 1 \text{ mC}$ moves at $v = 300 \text{ m s}^{-1}$ along x ; B along y ; E along z ; all perpendicular.

5(B)(a)

[2 marks]

State and explain the formula for the magnitude of the Lorentz force on a charge moving through a magnetic field. State the SI units.

ANSWER

$$F = BQv; F \text{ in N, } B \text{ in T, } Q \text{ in C, } v \text{ in m s}^{-1}$$

RULES YOU NEED

- $F = B Q v$ ($v \perp B$)

WORKING — STEP BY STEP

1 The formula

$$F = B Q v$$

2 Terms and units

F: force (N). B: magnetic field (T). Q: charge (C). v: velocity (m s^{-1}).

3 Explain

The force is perpendicular to both v and B . It pushes the charge sideways, so it changes its direction, not its speed.

Marks	Awarded for
1	$F = BQv$ ($v \perp B$).
1	All SI units.

5(B)(b)(i)**[3 marks]**

If $E = 10^3 \text{ V m}^{-1}$, what is the magnitude of the electric force on the particle? What is its direction?

ANSWER

$F_E = 1 \text{ N}$, along $+z$ (the direction of E)

RULES YOU NEED

- $F = q E$
- Positive charge: force along E
- $1 \text{ mC} = 10^{-3} \text{ C}$

WORKING — STEP BY STEP**1 Convert the charge**

$$q = 1 \text{ mC} = 10^{-3} \text{ C}$$

2 Electric force

$$F_E = q E = 10^{-3} \times 10^3 = 1 \text{ N}$$

3 Direction

The charge is positive, so the force points the same way as E : along $+z$.

Marks	Awarded for
1	$F = qE$.
1	1 N.
1	Along $+z$.

5(B)(b)(ii)**[3 marks]**

If $B = 2 \text{ T}$, what is the magnitude of the magnetic force on the particle? What is its direction?

ANSWER

$$\mathbf{F_B = 0.6 \text{ N, along } +z}$$

RULES YOU NEED

- $F = B q v$
- Force $\perp$ to both v and B (Left Hand Rule)

WORKING — STEP BY STEP**1 Magnitude**

$$F_B = 2 \times 10^{-3} \times 300 = 0.6 \text{ N}$$

2 Which axis?

v is along x and B along y . The force must be perpendicular to both, so it lies along the z -axis.

3 Which way along z ?

Left Hand Rule for a positive charge (current along $+x$, field along $+y$): the force is along $+z$, the same way as E .

Marks	Awarded for
1	$F = Bqv$.
1	0.6 N.
1	Along $+z$ (perpendicular to v and B).

5(B)(b)(iii)**[2 marks]**

Find the magnitude of the net force acting on the particle.

ANSWER

$F_{\text{net}} = 1.6 \text{ N (along +z)}$

RULES YOU NEED

- Forces in the same direction add

WORKING — STEP BY STEP**1 Both forces point along +z**

$$F_{\text{net}} = F_E + F_B = 1 + 0.6 = 1.6 \text{ N}$$

Marks	Awarded for
1	Both forces along the same axis, added.
1	1.6 N (own-figure with consistent directions accepted).

Question 5(B) Total: 10 Marks

Question 6 · Option A

Laws of reflection, total internal reflection, limit on a refractive index — 10 marks

Given: Glass $n = 1.5$ on a second transparent medium; ray meets the boundary at 50° and is totally internally reflected.

6(A)(a)

[2 marks]

Write down the two laws of reflection.

ANSWER

$\theta_i = \theta_r$; incident ray, reflected ray and normal are coplanar

RULES YOU NEED

- Angles are measured from the normal

WORKING — STEP BY STEP

1 Law 1

The angle of incidence equals the angle of reflection: $\theta_i = \theta_r$.

2 Law 2

The incident ray, the reflected ray and the normal all lie in the same plane.

Marks	Awarded for
1	$\theta_i = \theta_r$.
1	Same plane.

6(A)(b)**[3 marks]**

Using a diagram, describe total internal reflection. Derive an expression for the critical angle θ_{crit} , describing any other quantities.

ANSWER

$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$, with $n_1 > n_2$; above it, all light is reflected back

RULES YOU NEED

- Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- $\sin 90^\circ = 1$

WORKING — STEP BY STEP**1 Light going from dense to less dense**

From a higher n_1 to a lower n_2 (e.g. glass to air), the ray bends away from the normal.

2 Increase the angle

At the critical angle the refracted ray runs along the boundary (90°). Beyond it, no light escapes: it is all reflected back. That is **total internal reflection**.

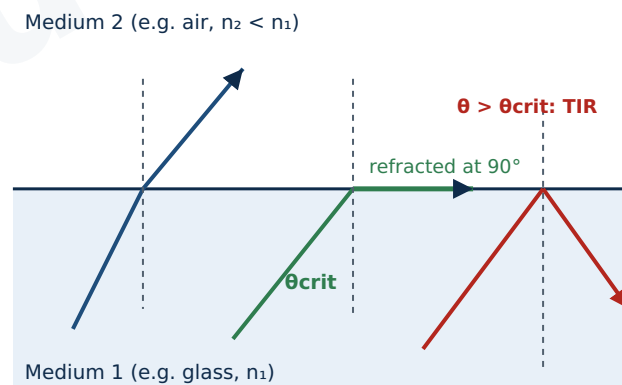


Fig. 2 — Below θ_{crit} the ray refracts out; at θ_{crit} it runs along the boundary; above θ_{crit} it is totally internally reflected.

3 Derive θ_{crit}

Put $\theta_2 = 90^\circ$ into Snell's law:

$$n_1 \sin \theta_{\text{crit}} = n_2 \times 1 \Rightarrow \theta_{\text{crit}} = \sin^{-1}(n_2 / n_1)$$

n_1, n_2 : refractive indices of the two media, with $n_1 > n_2$.

Marks	Awarded for
1	Diagram and description of TIR.
1	Derivation from Snell's law with $\theta_2 = 90^\circ$.
1	$\theta_{\text{crit}} = \sin^{-1}(n_2/n_1)$, terms described.

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6(A)(c)**[5 marks]**

Glass ($n = 1.5$) rests on a second medium. A ray meets it at 50° and undergoes total internal reflection. Calculate a limit on the refractive index of the second medium.

ANSWER

$$n_2 < 1.15$$

RULES YOU NEED

- TIR happens when $\theta > \theta_{\text{crit}}$
- $\sin \theta_{\text{crit}} = n_2 / n_1$

WORKING — STEP BY STEP**1 Condition for TIR**

The ray is totally reflected, so 50° must be bigger than the critical angle:

$$50^\circ > \theta_{\text{crit}}$$

2 Turn it into sines

Sine gets bigger as the angle gets bigger (up to 90°), so:

$$\sin 50^\circ > \sin \theta_{\text{crit}} = n_2 / 1.5$$

3 Rearrange for n_2

$$n_2 < 1.5 \times \sin 50^\circ$$

4 Calculate

$$n_2 < 1.5 \times 0.766 = 1.15$$

5 Interpret

This is an **upper** limit: the second medium's refractive index must be below 1.15 (TIR also needs $n_2 < n_1 = 1.5$, which this satisfies).

Marks	Awarded for
1	TIR condition: $\theta > \theta_{\text{crit}}$.
1	$\sin \theta_{\text{crit}} = n_2/n_1$.
1	Inequality set up correctly: $\sin 50^\circ > n_2/1.5$.
1	$n_2 < 1.5 \sin 50^\circ$.
1	$n_2 < 1.15$ (an upper limit).

Question 6(A) Total: 10 Marks

Question 6 · Option B

Huygens' principle, Young's double slit, interference — 10 marks

Given: Red laser $\lambda = 650 \text{ nm}$; slit separation $3.73 \mu\text{m}$; point P at 5° to the beam.

6(B)(a)

[2 marks]

Describe how Huygens' Principle can be used to describe the propagation of a wave as it meets an obstacle.

ANSWER

Each point on a wavefront is a source of wavelets; their envelope is the new wavefront, so waves spread around obstacles (diffraction)

RULES YOU NEED

- Each point on a wavefront is a source of new wavelets

WORKING — STEP BY STEP

1 The principle

Every point on a wavefront acts as a source of small spherical wavelets. The new wavefront is the surface that touches all these wavelets.

2 At an obstacle

Only the points that are not blocked make wavelets. These spread into the region behind the obstacle, so the wave bends around edges: **diffraction**.

Note on the lecture material: Huygens' principle is **not covered in the PY1010 lecture notes**. This answer is included for completeness.

Marks	Awarded for
1	Each point on a wavefront is a source of wavelets.
1	Applied to an obstacle: the wave spreads/bends around it.

6(B)(b)**[3 marks]**

Describe, with a diagram, Young's double slit experiment.

ANSWER

Two in-phase slit sources superpose to give bright (constructive) and dark (destructive) fringes on a screen

RULES YOU NEED

- Superposition: in phase → constructive; exactly out of phase → destructive

WORKING — STEP BY STEP**1 The set-up**

Light of one wavelength shines on two narrow slits close together. The slits act as two sources that are **in phase**.

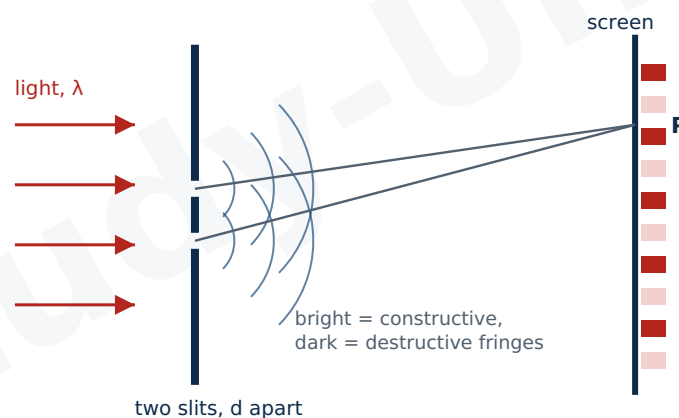


Fig. 3 — Two in-phase slits; the waves overlap and give bright and dark fringes on a screen.

2 What happens on the screen

The two waves overlap and superpose. Where they arrive in phase: constructive interference, a **bright** fringe. Where exactly out of phase: destructive, a **dark** fringe.

3 Why it matters

The alternating bright and dark bands show that light behaves as a wave.

Note on the lecture material: Young's double slit experiment is **not covered in the lecture notes**. The superposition and interference ideas used are, but the experiment itself comes from outside the lecture material.

Marks	Awarded for
1	Labelled diagram: source, two slits, screen.
1	Two coherent (in-phase) sources superpose.
1	Bright/dark fringes explained by constructive/destructive interference.

6(B)(b)(i)**[1 mark]**

The difference between the waves arriving at P is 2λ . Constructive or destructive? Explain.

ANSWER

Constructive: 2λ is a whole number of wavelengths, so the waves are in phase

RULES YOU NEED

- A whole number of wavelengths → in phase → constructive

WORKING — STEP BY STEP**1 Is 2λ a whole number of wavelengths?**

Yes ($n = 2$). Crest meets crest, so the waves arrive in phase and add: **constructive**.

Marks	Awarded for
1	Constructive, with reason.

6(B)(b)(ii)**[1 mark]**

If the difference is $\frac{3}{2} \lambda$, what type of interference is observed? Explain.

ANSWER

Destructive: 1.5λ puts the waves exactly out of phase

RULES YOU NEED

- An odd number of half-wavelengths → exactly out of phase → destructive

WORKING — STEP BY STEP**1 Is 1.5λ a half-number of wavelengths?**

Yes. Crest meets trough, so the waves are exactly out of phase and cancel:
destructive.

Marks**Awarded for****1**

Destructive, with reason.

6(B)(c)(i)**[2 marks]**

(Printed as a second “(b)” on the paper.) The line from the slits to P makes 5° with the beam. Calculate the path difference at P, in terms of λ .

ANSWER

Path difference = 0.5λ ($\lambda/2$)

RULES YOU NEED

- Path difference = $d \sin \theta$
- $1 \mu\text{m} = 10^{-6} \text{ m}$; $1 \text{ nm} = 10^{-9} \text{ m}$

WORKING — STEP BY STEP**1 Path difference formula**

The wave from the further slit travels an extra $d \sin \theta$:

$$\Delta = d \sin \theta$$

2 Put the numbers in

$$\Delta = 3.73 \times 10^{-6} \times \sin 5^\circ = 3.73 \times 10^{-6} \times 0.0872 = 3.25 \times 10^{-7} \text{ m}$$

3 Divide by the wavelength

$$\Delta / \lambda = 3.25 \times 10^{-7} \div 6.50 \times 10^{-7} = 0.50$$

Note on the lecture material: The path-difference formula $d \sin \theta$ is not in the lecture notes (it belongs with Young's experiment above).

Marks	Awarded for
1	$\Delta = d \sin \theta$.
1	0.5λ .

6(B)(c)(ii)**[1 mark]**

What type of interference do you expect in this case?

ANSWER**Destructive interference (dark fringe)****RULES YOU NEED**

- Half a wavelength → exactly out of phase

WORKING — STEP BY STEP**1 Apply the result from (i)**

0.5λ means crest meets trough: **destructive** interference, a dark fringe at P.

Marks	Awarded for
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1	Destructive.
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Question 6(B) Total: 10 Marks